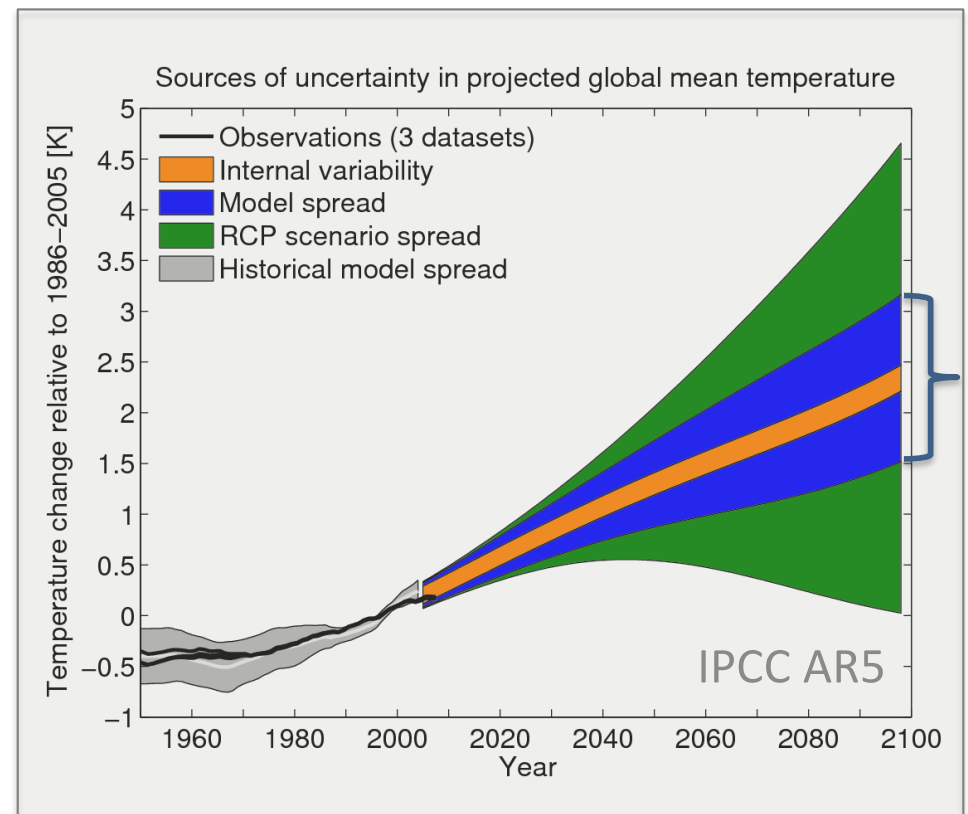
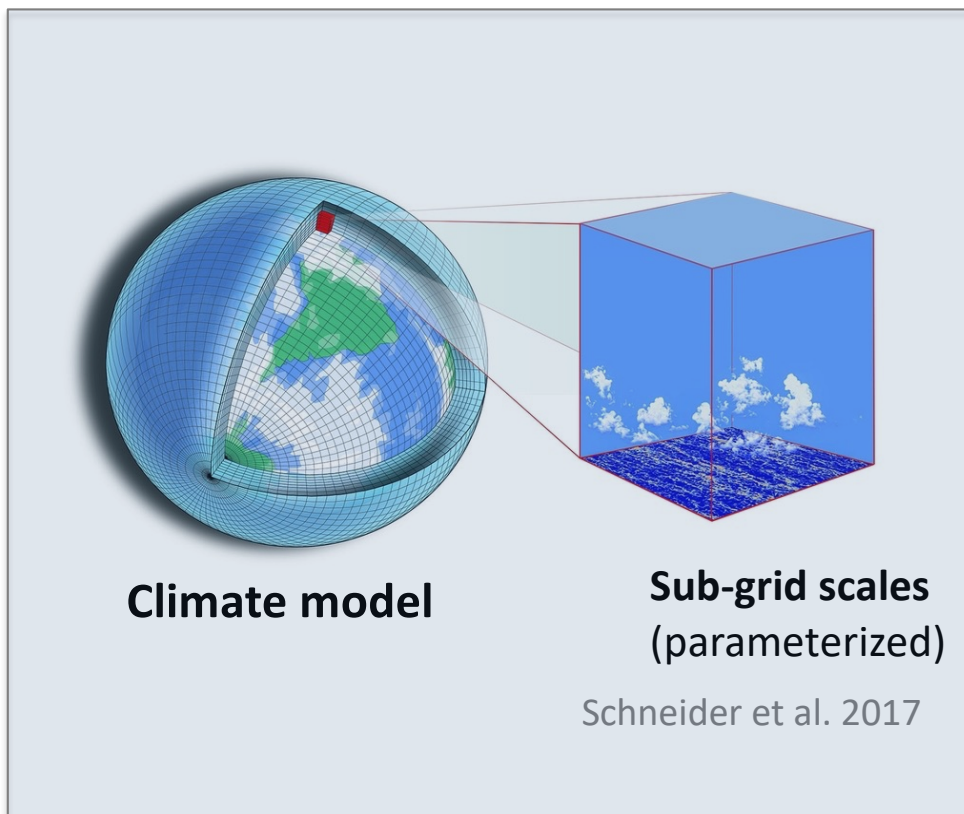


Topic 4: Sensitivity of equilibrium climate to physical parameterizations

Background

- Imperfect parameterizations of physical processes are responsible for the bulk of uncertainty in future climate projections. Therefore, improving climate simulations requires improving parameterizations.



Q: How to extract the impact of different model parameterizations on the simulated climate?

A: By running lots of climate simulations using different combinations of parameterizations.

Topic 4: Sensitivity of equilibrium climate to physical parameterizations

Numerical experiments

- **Setup:** simplified aqua-planet simulations with WRF model:
 - Earth-like planet, ocean only with prescribed sea surface temperature
 - No annual cycle/seasons, retained diurnal cycle
 - Zonally-symmetric forcings and (in a statistical sense) results
 - Two sets of simulations: (1) '**present**' - **control climate**, (2) '**climate change**' - **uniform +4K increase of SST**
- **Data:**
 - 2 x 8 simulations
 - 2 climates (control/+4K SST) and 8 different combinations of parameterizations
 - 12-monthly averaged data
 - On Jupyter Server data is in `/home/jovyan/shared/data/wrf/control` and `/home/jovyan/shared/data/wrf/4K`
- **Variables included:**
 - Horizontal and vertical velocity components
 - Cloud cover
 - Water vapor and temperature profiles
 - Precipitation
 - Radiative fluxes at the top of the atmosphere

Topic 4: Sensitivity of equilibrium climate to physical parameterizations

List of aqua-planet simulations

Experiment name	Parameterization			
	1	2	3	4
	Microphysics	Convection	Boundary layer	Radiation
St	WSM-6	GFS (AS)	MYNN-3	RRTMG
MPwdm	WDM-6			
MPmor	Morrison			
COkf		Kain-Frisch		
COTie		Tiedtke		
PBLnn2			MYNN, level 2	
PBLysu			YSU	
RAcam				CAM

Differences in the sets of parameterizations with respect to **St** experiment

Column of a climate model
(subgrid processes)



Reference: Cesana, G., Suselj, K., & Brient, F. (2017). On the dependence of cloud feedbacks on physical parameterizations in WRF aquaplanet simulations. *GRL*, 44. doi: 10.1002/2017GL074820

Topic 4: Sensitivity of equilibrium climate to physical parameterizations

Example questions to be answered

- Is aqua-planet climate a good approximation of the Earth's climate?
- Quantify the role of physical parameterizations for the control climate:
 - Which climate zones and physical quantities are most sensitive to physical parameterizations?
 - Which physical parameterizations control that variability?
- Investigate the role of physical parameterizations in the simulated “climate change”:
 - How do atmospheric properties change with climate change (from control to +4K experiments)?
 - How sensitive are these changes to different physical parameterizations?

Example results (1): Is aqua-planet climate realistic?

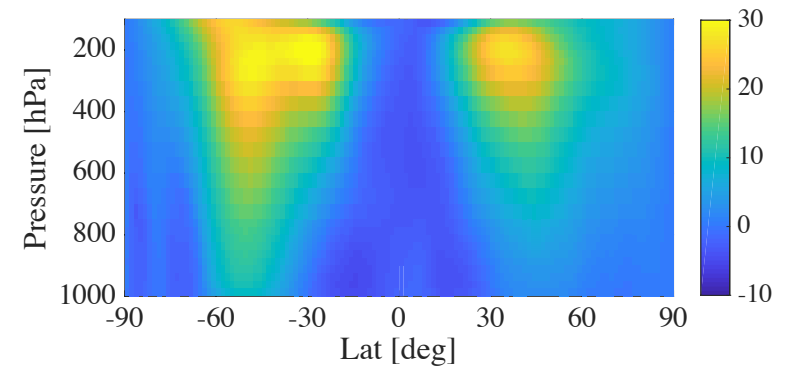
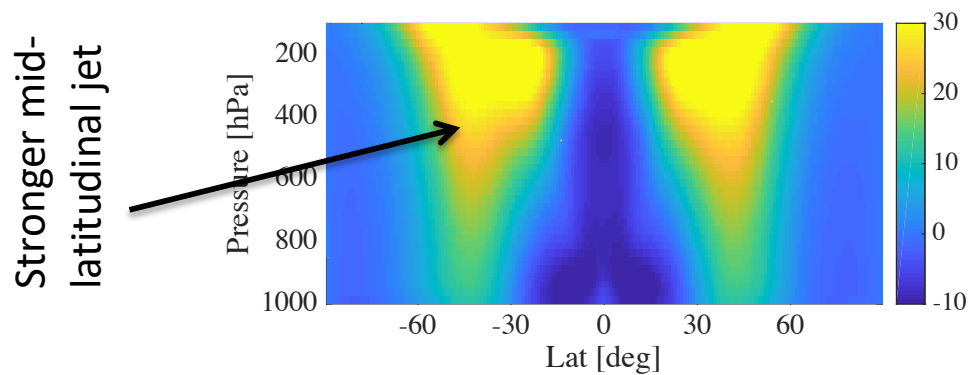
Hypothesis: Main features of the aqua-planet climate are comparable to real climate

Methodology: compare zonally-averaged fields from the St experiment and reanalysis

Aqua-planet: WRF/St
(simplified world)

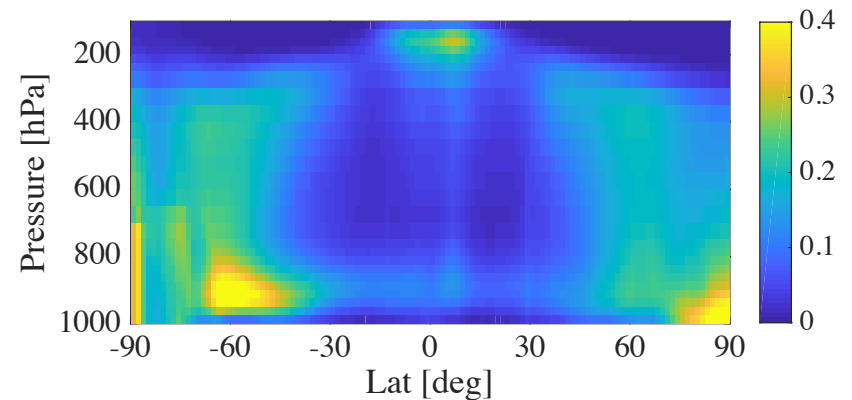
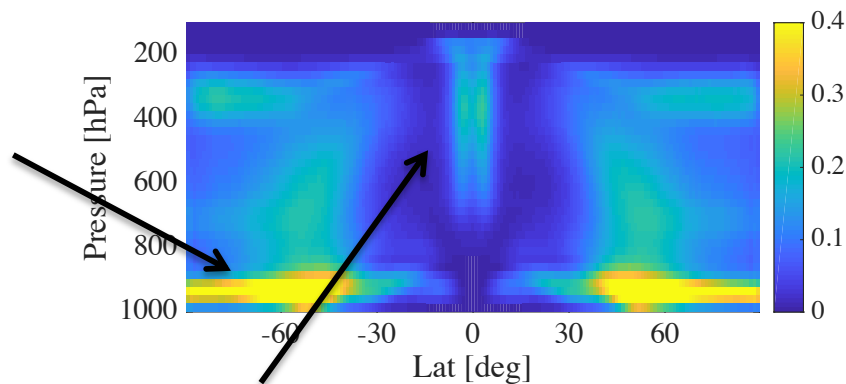
Zonal wind [m s^{-1}]

Reanalysis (ECMWF/Interim)
("real world")



Higher near-surface cloudiness

Cloud fraction



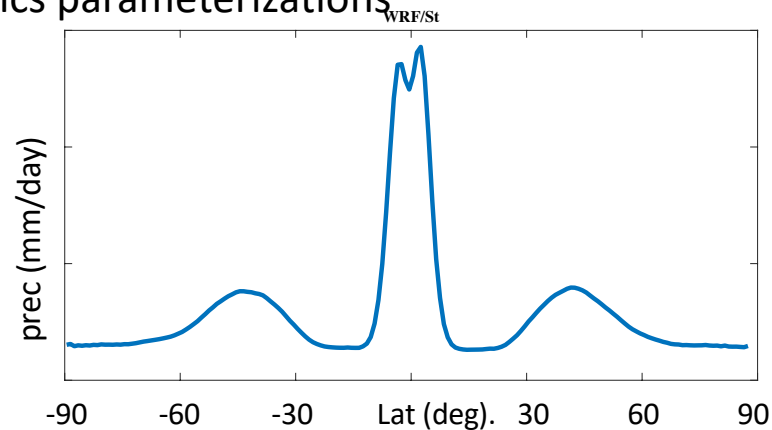
More tropical mid-level clouds

Example results (2): Sensitivity of climate to physical parameterizations

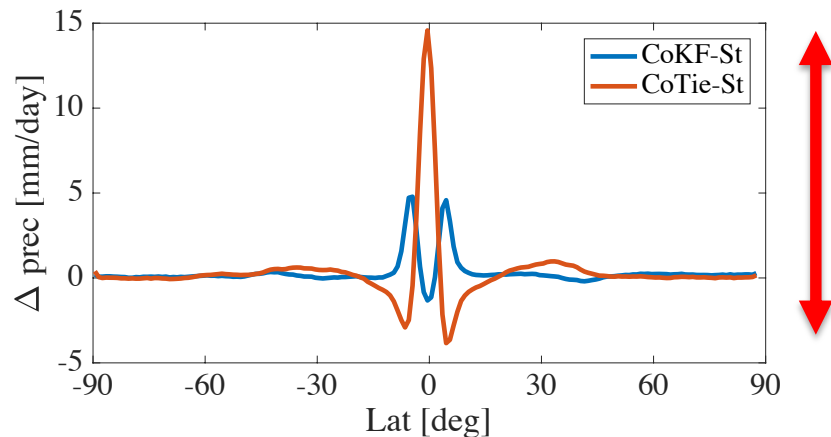
Hypothesis: Convective and microphysics parameterizations determine the dynamics, cloud structure and precipitation patterns in the tropics

Methodology: Compare precipitation variability for the control climate due to convection and microphysics parameterizations

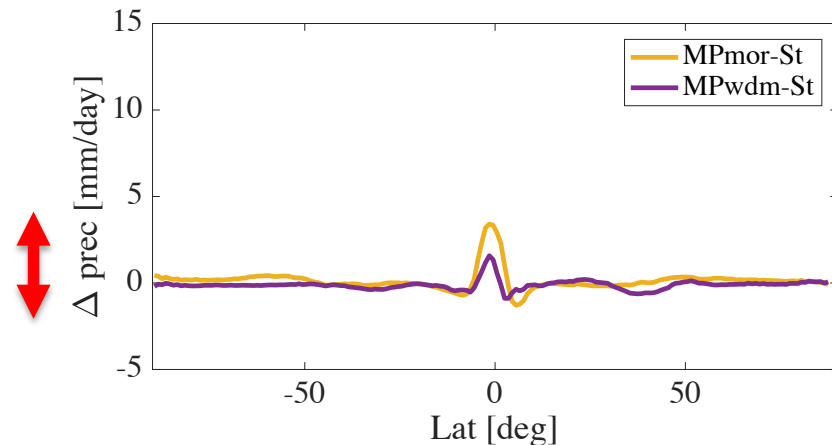
Zonally-averaged surface precipitation from WRF/St



Differences in surface rain rate - convection
(2 parameterizations from WRF/St)



Differences in surface rain rate – microphysics
(2 parameterizations from WRF/St)

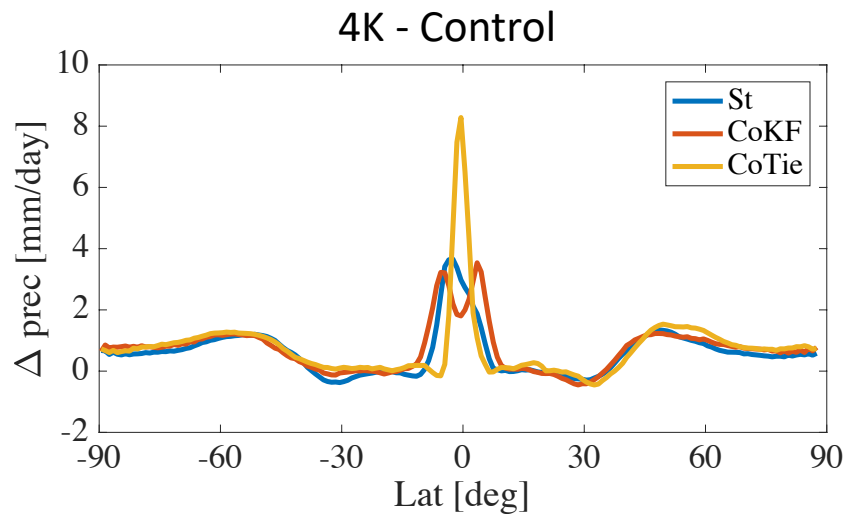


Example results (3): Sensitivity of climate response to increased SST

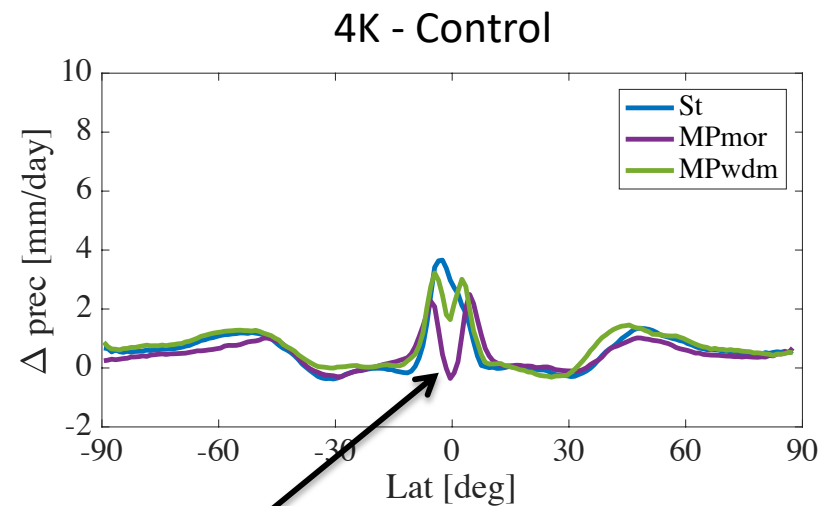
Hypothesis: Climate change uncertainty in the tropics is mostly sensitive to convection parameterizations

Methodology: Compare control and warm climate statistics in the tropics

Changes due to different convection parameterizations



Changes due to different microphysics parameterizations



Microphysical parameterizations strongly impact the variability of tropical rain changes to climate warming

Your task

- Familiarize yourself with the available data sets
- Study the simple examples provided above
- Formulate your own hypothesis
- Use the available data to support/reject your hypothesis
- Make a story based on your results

Good luck!

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